

Data Science Project – Prediction of User Quality of Experience (QoE)

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1 Context and Objectives

This project is based on a dataset describing mobile video streaming sessions. Each observation corresponds to a single session and is characterized by technical playback indicators, network information, device characteristics, user-related attributes, and a subjective Mean Opinion Score (MOS).

The dataset is composed of several groups of variables, each describing a different aspect of a video streaming session. The meaning of the columns is explained by grouping them into application-level (QoA), network-level (QoS), device-level (QoD), and user-level (QoU) features.

Within each group, variables represent either measured technical quantities (e.g. bitrate, buffering time) or categorical descriptors (e.g. network type, device model).

The analysis focuses on understanding how different categories of variables contribute to user experience, rather than on raw prediction alone.

The main objective of the project is to predict the MOS score as a multi-class classification problem. In addition to predictive performance, the analysis aims to:

- characterize the dataset through exploratory data analysis,
- justify preprocessing choices based on observed data properties,
- evaluate the contribution of different feature groups,
- and compare the behavior of several machine learning models on this task.

2 Dataset Description

The dataset contains several hundred mobile video streaming sessions, each described by a heterogeneous set of variables. Each row corresponds to a single streaming session, while the columns describe different aspects of that session.

For clarity, variables are grouped into four categories. Quality of Application (QoA) features describe technical playback conditions, such as bitrate, frame rate, buffering duration, and dropped frames.

Quality of Service (QoS) variables provide information about the network conditions under which the session took place.

Quality of Device (QoD) features characterize the user's device, including model and operating system information.

Finally, Quality of User (QoU) variables capture basic user-related context, such as age, gender, and usage conditions.

Some variables are numerical and represent measured quantities (e.g. buffering time, bitrate), while others are categorical and describe discrete attributes (e.g. network type, device model).

The target variable is the Mean Opinion Score (MOS), which takes integer values from 1 to 5. Lower values correspond to poor perceived quality, while higher values indicate better user experience.

3 Exploratory Data Analysis

3.1 Device Distribution

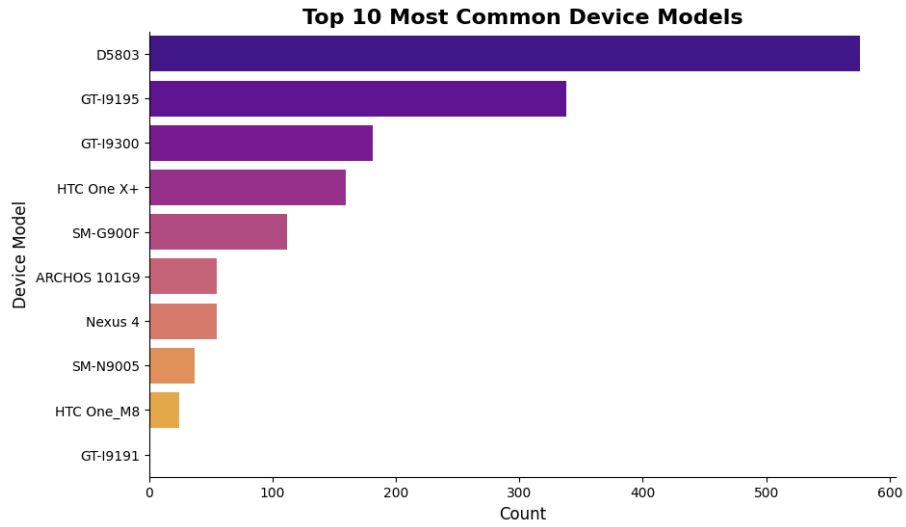


Figure 1: Most frequent device models

The distribution of device models is highly skewed. A small number of smartphone models account for a large proportion of the observations, while many other devices appear only rarely.

This concentration suggests that device-related variables may capture dataset-specific patterns rather than general effects on Quality of Experience. As a result, these variables risk introducing bias or overfitting, especially for models sensitive to categorical sparsity.

3.2 Target Variable Distribution

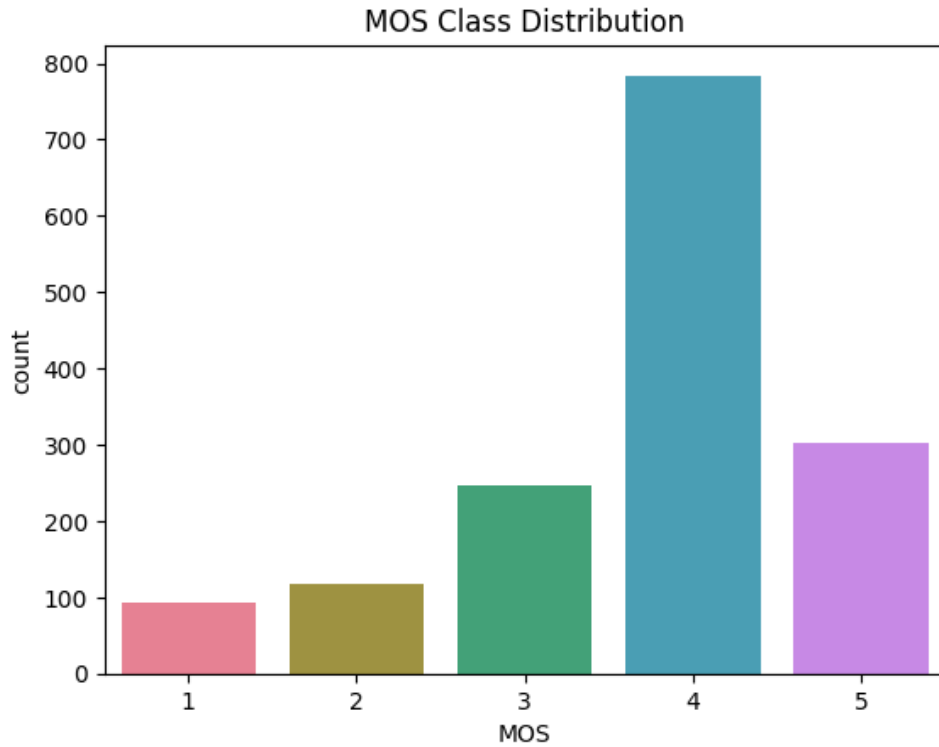


Figure 2: MOS class distribution

The MOS distribution is clearly imbalanced. High quality scores ($\text{MOS} = 4$ and 5) dominate the dataset, while low scores are relatively rare. This reflects realistic usage conditions, where most streaming sessions are acceptable or good.

However, this imbalance implies that naïve accuracy is not an appropriate evaluation metric. It also suggests that models may tend to over-predict higher MOS values unless corrective strategies are applied.

3.3 Univariate Analysis of Numerical Features

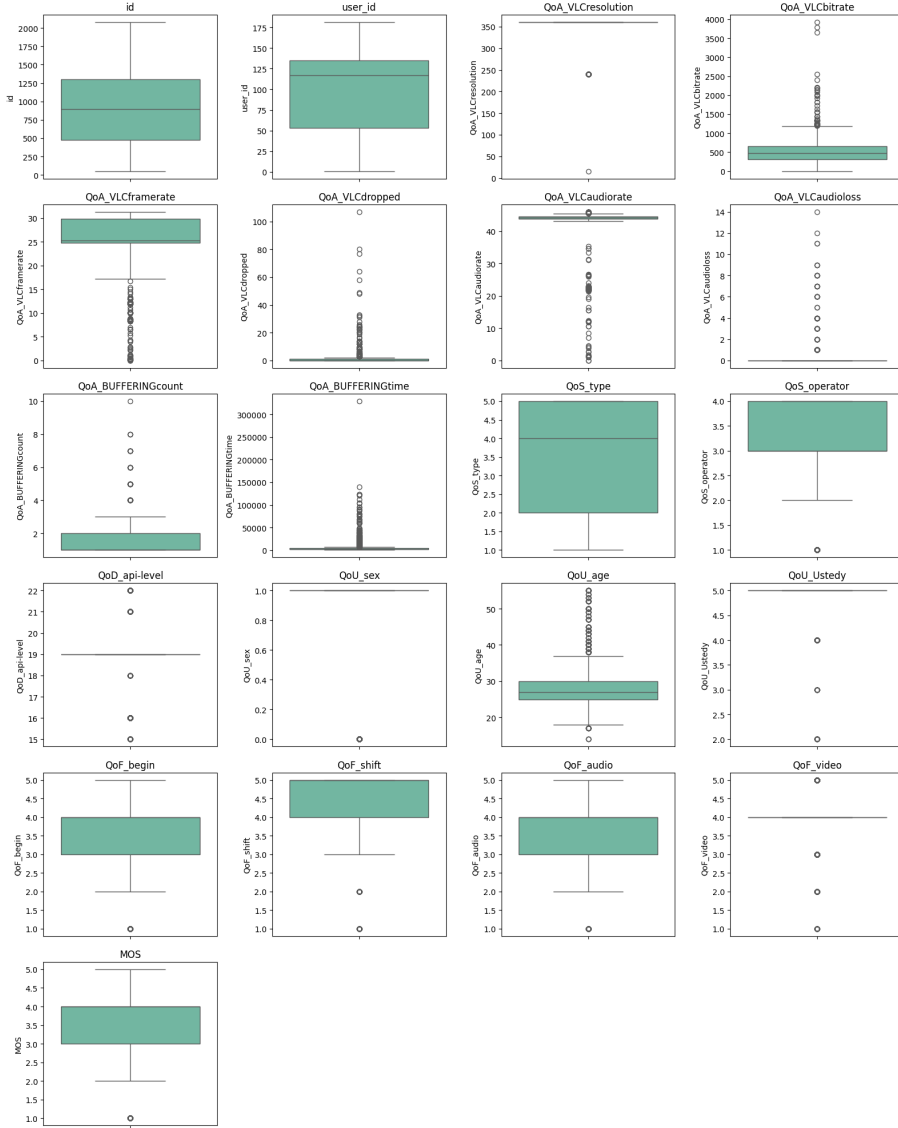


Figure 3: Boxplots of numerical features

The boxplots reveal strong heterogeneity in the numerical features, particularly for application-level QoA variables. Buffering time, buffering count, dropped frames, and audio loss exhibit highly skewed distributions with extreme outliers.

These extreme values correspond to rare but severe playback degradations. Rather than being errors, they represent critical situations that are likely to strongly affect perceived quality. This observation motivates preserving these samples while controlling their influence through robust preprocessing techniques.

3.4 Correlation Structure

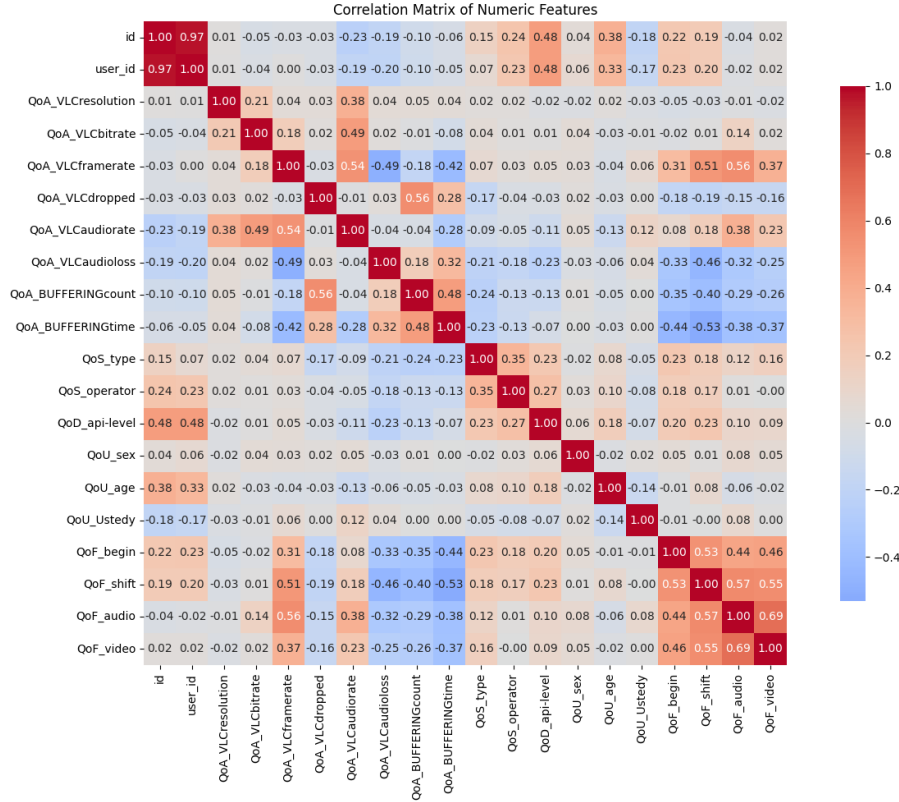


Figure 4: Correlation matrix of numerical features

The correlation matrix highlights several meaningful relationships. Buffering-related variables are moderately correlated with each other and with dropped frames, reflecting the underlying mechanics of video streaming.

Importantly, no extremely high correlations are observed. This suggests that while features are related, they capture complementary aspects of playback quality. As a result, no variables were removed solely based on correlation, especially given that tree-based models can naturally handle correlated inputs.

4 Data Science Protocol and Preprocessing

4.1 Feature Selection and Cleaning

Identifier variables (`id` and `user_id`) were removed prior to model training. These variables uniquely identify sessions or users but do not carry meaningful information about perceived quality. Keeping them could introduce noise or spurious correlations without improving generalization.

No missing or extreme values were detected in the dataset.

Correlation analysis did not reveal strongly redundant variables. As a result, no features were removed solely based on correlation, especially since tree-based models can naturally handle correlated inputs.

4.2 Train-Test Split

The dataset was split into training and test sets using an 80/20 ratio. The test set was kept completely separate and used only for final evaluation. All preprocessing steps, feature selection

choices, and hyperparameter tuning were performed using the training set only. This protocol ensures an unbiased evaluation of model generalization.

4.3 Feature Types and Encoding

Categorical variables were encoded using one-hot encoding. This choice avoids introducing artificial ordinal relationships and allows models to treat categories independently.

4.4 Feature Scaling

Numerical features were standardized using z-score normalization. This step is essential for models such as Logistic Regression, SVM, and KNN, which are sensitive to feature scale. Although tree-based models do not require scaling, a unified preprocessing pipeline was used for consistency and fair comparison.

4.5 Outlier Handling

Exploratory analysis highlighted the presence of extreme values in buffering time, buffering count, dropped frames, and audio loss. These values correspond to rare but meaningful degradation events.

Instead of removing observations, a winsorisation strategy was applied at the 1st and 99th percentiles. This approach preserves all samples while limiting the influence of extreme values on models sensitive to large magnitudes.

4.6 Feature Engineering

A derived feature combining buffering duration and bitrate was initially included in the full feature set. The goal was to explicitly capture the interaction between throughput and playback interruptions.

However, when experimenting with reduced feature sets, this engineered feature was removed together with QoD and QoU variables to maintain a consistent and interpretable comparison.

4.7 Alternative Feature Sets

Two main feature configurations were evaluated:

- the full feature set,
- a reduced feature set excluding QoD and QoU variables.

Interestingly, removing device- and user-related variables led to a slight improvement in performance. This suggests that these variables introduce noise or dataset-specific biases rather than robust predictive information.

4.8 Class Imbalance Considerations

Given the imbalance in MOS classes, class-weighted loss functions were used for models that support them.

In addition model performance was evaluated primarily using the weighted F1-score rather than accuracy.

This strategy ensures that minority classes contribute proportionally to the training objective.

5 Modeling Strategy

The dataset used in this project is fully tabular and contains a mix of numerical and categorical variables. For this reason, the modeling strategy focused on classical machine learning models that are well suited for structured data, rather than very complex architectures that typically require much larger datasets.

The objective was not to maximize model complexity, but to compare different modeling approaches and understand which assumptions are most appropriate for this type of data.

This also makes it easier to interpret the results and relate model behavior back to meaningful QoE-related factor.

5.1 Problem Formulation

Even though MOS has an ordinal meaning, it was treated as a multi-class classification problem. This choice makes it easier to analyse prediction errors between quality levels and avoids assuming that the distance between two consecutive scores is always the same.

5.2 Models Considered

Several models were tested in order to cover different levels of complexity and different learning mechanisms:

- **Logistic Regression**, used as a simple baseline to evaluate how much information can be captured using linear relationships only.
- **Support Vector Machine (SVM)**, allowing non-linear decision boundaries.
- **K-Nearest Neighbors (KNN)**,
- **Decision Tree and Random Forest**, Since it can naturally capture non-linear interactions and are robust to outliers.
- **Gradient Boosting / XGBoost**,
- **Multi-Layer Perceptron (MLP)**, tested to check whether a neural network could outperform classical models on this dataset.

All models were trained using the same preprocessing pipeline in order to ensure a fair comparison.

5.3 Evaluation Metrics

Because the MOS classes are not equally represented, accuracy alone is not sufficient to evaluate performance. The weighted F1-score was therefore used as the main evaluation metric, as it accounts for both precision and recall while giving more importance to frequent classes.

In addition, confusion matrices were analysed for all models in order to better understand which MOS values were most often confused.

5.4 Hyperparameter Selection

For each model, a limited grid search combined with cross-validation was used to select reasonable hyperparameter values. This approach allowed performance to be improved without overfitting or excessive tuning.

All model selection steps were performed using the training set only, and the test set was kept strictly for final evaluation.

6 Results and Analysis

6.1 Overall Performance

	Model	Accuracy	F1_weighted	F1_macro
3	Random Forest (tuned)	0.828479	0.832115	0.773425
4	XGBoost (tuned)	0.822006	0.820844	0.782551
0	Logistic Regression	0.792880	0.796615	0.734760
1	SVM	0.796117	0.792304	0.708326
5	MLP	0.776699	0.775094	0.703193
2	KNN	0.724919	0.702590	0.632362

Figure 5: Performance comparison across models

Tree-based ensemble models achieve the strongest overall performance. Random Forest obtains the highest weighted F1-score, closely followed by XGBoost. This indicates that non-linear interactions between playback-related variables are essential to accurately predict perceived quality.

Linear and distance-based models perform reasonably but consistently below ensemble methods. KNN is particularly affected by class imbalance and noisy dimensions, which limits its discriminative power.

The neural network does not outperform tree-based models. This result is expected given the limited dataset size and the dominance of structured, tabular features.

6.2 Confusion Matrix Analysis



Figure 6: Confusion matrices for all models

Confusion matrices provide a detailed view of model behavior. Across all models, errors occur predominantly between adjacent MOS levels, especially between scores 3 and 4.

This pattern reflects the subjective nature of quality assessment: sessions near the boundary between acceptable and good quality are inherently ambiguous. In contrast, extreme confusions (e.g., MOS 1 predicted as MOS 5) are rare, indicating that models reliably distinguish very poor from very good experiences.

Tree-based models show sharper diagonals and fewer off-diagonal errors, confirming their superior ability to separate quality levels.

6.3 Feature Importance and SHAP Analysis

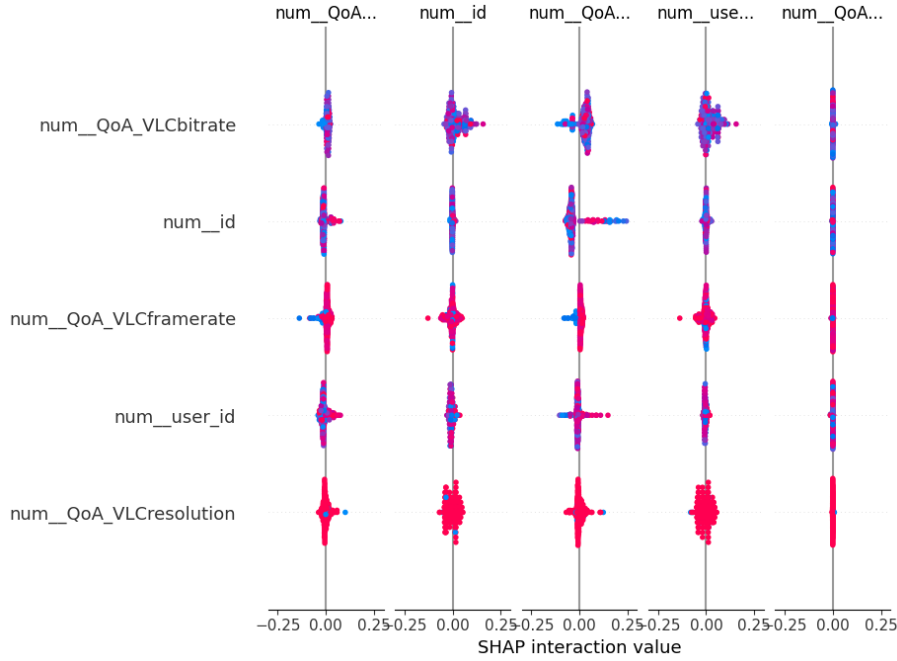


Figure 7: SHAP analysis of the Random Forest model

SHAP analysis highlights the dominant influence of application-level QoA variables. Buffering duration, buffering count, dropped frames, and bitrate consistently contribute the most to model predictions.

High buffering time and frequent interruptions strongly decrease predicted MOS, while higher bitrate and stable playback increase it. Device-related and user-related features have much weaker and less consistent contributions.

SHAP interaction plots further indicate that buffering effects are amplified under low bitrate conditions, confirming that perceived quality depends on combined playback factors rather than isolated metrics.

6.4 Ablation Studies

	Model	Accuracy	F1_weighted	F1_macro
3	Random Forest (tuned)	0.834951	0.839436	0.796735
4	XGBoost (tuned)	0.822006	0.822204	0.798501
1	SVM	0.822006	0.821704	0.765619
0	Logistic Regression	0.805825	0.809529	0.759282
5	MLP	0.779935	0.776356	0.689933
2	KNN	0.750809	0.743797	0.655191

Figure 8: Performance comparison for reduced feature sets

Removing device-related (QoD) and user-related (QoU) variables results in a slight improvement in performance. This indicates that these variables introduce noise or dataset-specific bias rather

than useful predictive information.

The engineered feature combining buffering duration and bitrate was present in the initial full feature set. When QoD and QoU variables were removed, this engineered feature was also discarded. Its removal did not degrade performance, suggesting that tree-based models are already able to learn such interactions directly from the original QoA variables.



Figure 9: Confusion matrices using reduced feature sets

Confusion matrices for the reduced feature set show slightly improved concentration along the diagonal, particularly for ensemble models. This confirms that removing weakly informative variables improves generalization rather than harming it.

7 Discussion and Limitations

A first important observation is the clear dominance of application-level QoA features. Across all models, buffering-related variables (buffering time, buffering count, dropped frames) consistently appeared as the most influential factors. This is pretty intuitive since interruptions and playback instability are immediately perceived by users and strongly impact their experience. In contrast, device-related (QoD) and user-related (QoU) variables had a much weaker influence. In fact, removing these features slightly improved performance, suggesting that they may introduce noise rather than useful information in this dataset. This does not mean that device or user characteristics are irrelevant in general, but rather that, in this specific context, their effect is largely dominated by playback conditions.

Another key point concerns the behavior of different model families. Linear and distance-based models such as Logistic Regression and KNN achieved reasonable performance but struggled to capture more complex relationships between features. Tree-based ensemble methods, especially Random Forest and XGBoost, consistently outperformed the other approaches. Their

ability to model non-linear interactions and handle heterogeneous features without heavy pre-processing makes them particularly well suited for this type of tabular data. The neural network, despite additional tuning, did not outperform these models, which is likely due to the limited dataset size and the structured nature of the features.

The confusion matrix analysis provides additional insight beyond aggregate metrics. Most prediction errors occur between adjacent MOS levels, especially between scores 3 and 4. This is an important observation, as it reflects the inherent subjectivity of quality assessment. Distinguishing between medium and good quality is naturally more difficult than separating very poor from excellent experiences. The rarity of extreme misclassifications indicates that the models generally capture the overall quality level correctly, even when they fail to predict the exact score.

Feature importance and SHAP analysis further support these conclusions. They confirm that buffering and bitrate-related variables drive most predictions, while user demographics and device information have limited impact. The SHAP interaction analysis also suggests that some non-linear relationships exist between bitrate, buffering, and playback stability, which explains why ensemble models perform particularly well.

Despite these positive results, several limitations must be acknowledged. First, the dataset size remains relatively small, which limits the potential benefits of more complex models such as deep neural networks. Second, MOS is a subjective metric by nature, and individual user perceptions may vary even under similar technical conditions. Such variability cannot be fully captured by objective features alone. Finally, the dataset focuses on a specific use case (mobile video streaming), and the conclusions may not directly generalize to other multimedia services or network environments without further validation.